

TrES-3b

R-2572

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110324 · created 2026-08-27T00:01:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455644.92281 +/- 0.00085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.021
Transit depth (Rp/Rs)^2	2.49 +/- 0.65 [%]
Orbital Inclination (inc)	82.33 +/- 0.91
Airmass coefficient 1 (a1)	1.257 +/- 0.012
Airmass coefficient 2 (a2)	-0.2004 +/- 0.0063
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	None
Optimal Aperture	3.53
Optimal Annulus	10.99
Transit Duration (day)	0.0575 +/- 0.0089

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.158 $\pm$ 0.021 vs 0.16309 (deviation 0.24 σ)
Duration fit vs published	0.0575 d vs 0.0567 d (deviation 0.09 σ)
Mid-time O–C	1.6 min
Depth SNR	7.5
Residual scatter	0.94 %

Light curve

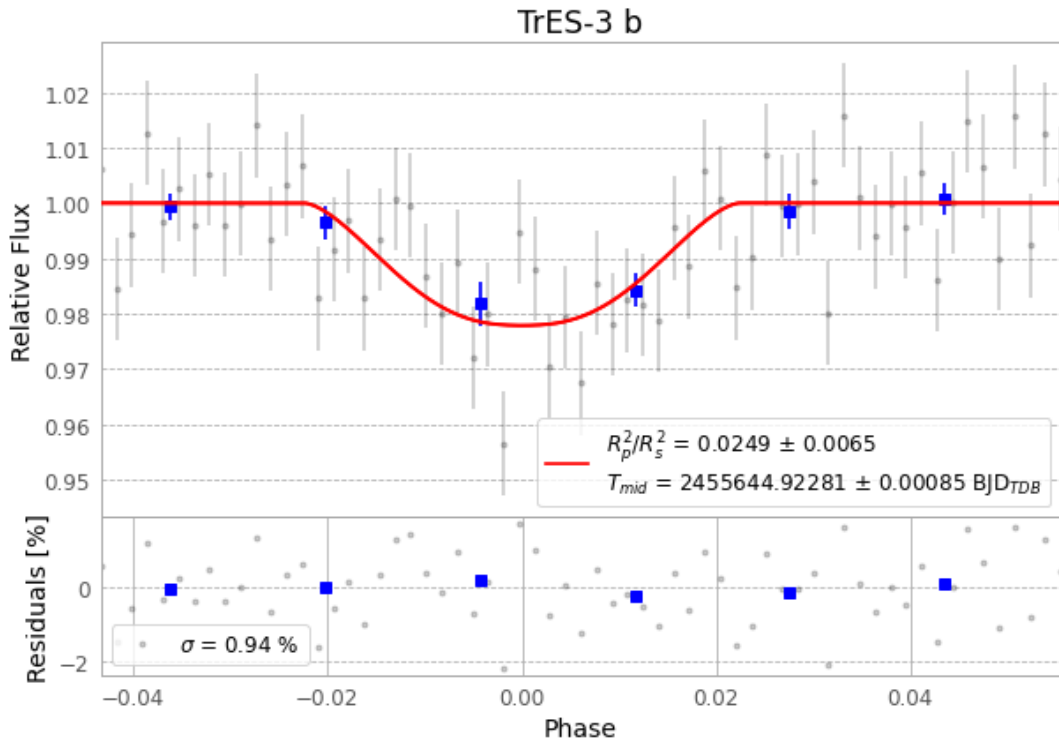


Figure 1 — FinalLightCurve_TrES-3 b_2011-03-24.png (SHA-256 d42d33684a6b0a0c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [243, 181], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 5f20b1e134c0b753...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), TRES-3110324084513.FITS → TRES-3110324115113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110324.log (6c91c0aed8f3...), FinalLightCurve_TrES-3 b_2011-03-24.png (d42d33684a6b...), FinalLightCurve_TrES-3 b_2011-03-24.csv (a32fc208e998...)

Compute resources

Wall time 102.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 00:01Z.